

CBSE Maths Questions 2015

SECTION A

1. If the quadratic equation $px^2 - 2\sqrt{5}px + 15 = 0$ has two equal roots, then find the value of p .
2. In Figure 1.2, a tower AB is 20 m high and BC, its shadow on the ground, is 203 m long. Find the Sun's altitude.

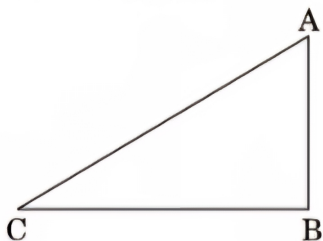


Fig. 1.2.

Figure 1: Figure 1.2

3. Two different dice are tossed together. Find the probability that the product of the two numbers on the top of the dice is 6.
4. In Figure, PQ is a chord of a circle with centre O and PT is a tangent. If $\angle QPT = 60^\circ$, find $\angle PRQ$.

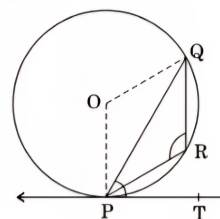


Fig. 1.4.

Figure 2: Figure 1.4

5. In Figure 1.5, two tangents RQ and RP are drawn from an external point R to the circle with centre O. If $\angle PRQ = 120^\circ$, then prove that $OR = PR + RQ$.

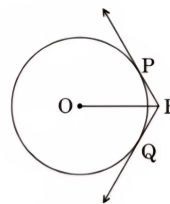


Fig. 1.5.

Figure 3: Figure 1.5

SECTION B

1. In Figure 2.1, a triangle ABC is drawn to circumscribe a circle of radius 3 cm, such that the segments BD and DC are respectively of lengths 6 cm and 9 cm. If the area of $\triangle ABC = 54\text{cm}^2$, find the lengths of sides AB and AC.

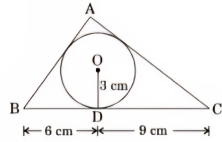


Fig. 2.1.

Figure 4: Figure 2.1

2. Solve the quadratic equation for x :
 $4x^2 + 4bx - (a^2 - b^2) = 0$
3. In an AP, if $S_5 + S_7 = 167$ and $S_{10} = 235$, then find the AP, where S_n denotes the sum of the first n terms.

4. The points $A = (4, 7)$, $B = (p, 3)$, and $C = (7, 3)$ are vertices of a right triangle, right-angled at B. Find the value of p .
5. Find the relation between x and y if the points $A = (x, y)$, $B = (-5, 7)$, and $C = (-4, 5)$ are collinear.

SECTION C

1. The 14th term of an AP is twice its 8th term. If its 6th term is -8, then find the sum of the first 20 terms.